

Forms as Boundary Genres in Medicine, Science, and Business

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This article examines medical treatment forms as boundary genres, drawing on genre and disciplinary studies theories to argue that medical forms represent a commingling of the business, science, and medical professions in ways that show evidence of tension and conflict between the disciplines.

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The disciplines characterize, classify, specialize; they distribute along a scale, around a norm, hierarchize individuals in relation to one another and, if necessary, disqualify and invalidate. (Foucault, 1975/1979, p. 223)

As Foucault (1975/1979) pointed out, relationships between disciplines are frequently characterized by competition, tension, and hierarchies. Although we often think of disciplines as coexisting peacefully across campuses, in which disciplinary experts agreeably respect and support each other, such a utopian picture obviously cannot be widespread. Moreover, if we accept the theory that disciplines experience tension in their relationships with each other, tensions that can be better understood by looking at the disciplines involved, we may begin to explain why certain tensions exist within our society. Health care has become an area rife with intense tension between health care providers, insurers, patients, and government agencies and committees. Tensions arise as patients seek out affordable yet effective health care, as physicians try to treat complex diseases with increasingly expensive technology, and as insurance companies struggle to find ways to minimize health-care costs so that they can continue to insure a growing population. The cause of this tension may be understood by exploring the extradisciplinary hierarchies of the medical profession as revealed in its genres.

Recognizing that disciplines and professions are not necessarily synonymous, I turn to studies of interdisciplinarity to understand the impact that other professions have on the profession of medicine. As a profession, medicine is segmented into fields and is also heavily influenced by other professions, such as science and business. In his study of medical fields, or specialties, Halpern (1992) found that relations between intrafield segments of a specialty influence its interfield boundaries. In other words, the hierarchical ways in which ancillary technicians and other skilled laborers are circumscribed within a specialty directly change how that specialty stands in regard to other specialties and to the overarching medical profession. Halpern further argued that disciplinary dominance depends on the subordination of one segment in cooperation with neighboring segments. Likewise, in what they call a chronotopic analysis, Detweiler and Peyton (1999) explored the temporal and spatial indicators that helped establish the success of occupational therapy as it moved from a subfield of medicine into a profession in its own right. Thus, we recognize that disciplinary or professional acknowledgment, success, and even dominance are important factors within the larger profession of medicine.

By combining a strong tradition in the inductive and deductive methods and theories of science, applying that knowledge in patient treatments, and performing such business practices as billing, filing, and managing staff, medicine is more than a separate, isolable profession; it is interdisciplinary. The extent to which these two professions—science and business—influence the profession of medicine remains a somewhat controversial subject of research. For example, Hunter (1991) argued forcefully that medicine, despite some popular perceptions to the contrary and “all its reliance on esoteric knowledge and sophisticated technology, is not a science” (p. xvii). Even though “the ideal of scientific rationality” is fundamental to our notions of the medical profession; of the complex relationships between biology, technology, and medicine; and of physicians as objective predictors of the physical world and our bodies, and medicine is dependent on sciences such as biology and chemistry, medicine is nevertheless “a practical body of knowledge [or applied knowledge] brought to bear on the understanding and treatment of particular cases” (Hunter, 1991, p. xviii).

Berg (1997), however, disagreed. His account of the post–World War II medical field emphasized the myriad of ways in which the public, the media, and even medicine itself came to view medicine as

a science or even sciences. He described medicine's increasing specialization, its increasing practice in such scientifically structured activities as prediction and diagnostic reasoning, its calls for standardized measures of vocabulary and treatment practices, and its almost mathematical reasoning (pp. 19-29). Berg acknowledged that although the perceptions of science change, since the 1950s, medicine has increasingly aligned itself with the scientific discipline. Moreover, as Berg noted, medicine also came to be viewed through the lens of the business sphere as the financial cost of health care became an issue, with editorials in medical journals in the 1970s and 1980s exhorting physicians to examine the cost of health-care treatments. And according to Berg, business-related methodologies, such as mathematical formulas, financial problem solving, and statistical analysis, were also clearly aligned with medical practice, as various authors in medical journals described medicine as a calculable process. This business-like alignment was surely influenced by the wave of health-care cost reforms that were legislated in the late 1960s and early 1970s, when the nation began to consider ways to limit and freeze the inflationary costs of health care (Starr, 1982, p. 399).

Despite uncertainty about the extent to which it is a science, medicine clearly draws heavily from scientific knowledge and predictive methods. Furthermore, during the past few decades, medicine has depended heavily on financial processes related to business decision making and has increasingly been called to account for the financial costs of health care. Undoubtedly then, medicine is influenced by the professions of science and business—their values in determining what is important, methodologies, popular image, and knowledge base. Given their coexistence with and influence on the profession of medicine, to what extent are these two disciplines or professions represented in medical documents? Furthermore, given Foucault's (1975/1979) theory that disciplines often compete for power against each other, how is that competition revealed or constructed in medical documents or genres?

Genres are regulated textual forms functioning in repeating situations. Schryer (2000) defined *genre* as both stabilized and improvisational, changing or maintaining as local needs and different communities dictate. She maintained that genres are "the constellations of regulated, improvisational strategies triggered by the interaction between individual socialization and an organization" (p. 9). And Miller (1984) suggested that genres are not merely fixed textual forms but represent social activity through textual regularities. Genres are

the means by which things get done within a community, ideas get transmitted, and plans get made. Miller argued that genres are “a form of social knowledge—a mutual construing of objects, events, interests, and purposes that not only links them but also makes them what they are: an objectified social deed” (p. 157). Both Miller and Schryer highlight the social element as they explain that genres embody the cultural rules, values, and motives of their respective communities, not the private intent of an individual. Bawarshi (2000) added that genres “help us define and organize kinds of social actions, social actions that these texts [genres] make possible” (p. 335). Although noting that genres work in social situations, Bawarshi made explicit his claim that genres work also on the situation. In other words, genres help construct that social situation. As they become more familiar to those people working in the situation, genres then constitute that situation, helping to

structure the way these individuals conceptualize and experience these situations, predicting their notions of what constitutes appropriate and possible responses and actions. This is why genres are both functional and epistemological—they help us function within particular situations at the same time they help shape the ways we come to know these situations. (p. 340)

Thus, genres shape how we work and the roles that we assume. This powerful nature of genres, then, becomes particularly important when we consider the ways in which disciplines and professions, such as science, medicine, and business, compete for power through the use of genres.

Other genre scholars, such as Swales (1990) and Berkenkotter (1993), have focused on the ways in which genres are produced by and serve the needs of single specific discourse communities. Swales found that a genre is the agreed-on product of a discourse community. Similarly, Berkenkotter claimed that a genre is dynamic in nature, always reflecting the shared conventions of the community as it changes. And Bazerman (1995) has analyzed the ways in which several interrelated genres, what he termed a *genre system*, work to meet the purposes of a community. But these early studies by Swales, Berkenkotter, and Bazerman as well as Miller (1984) do not examine the ways that genres frequently communicate across fields and disciplines, so we might question how a genre, when it is the agreed-on

product of a single community, could perhaps be understood, used, or prescribed by another community.

One way in which we can understand medical genres as multidisciplinary or multiprofessional is to study these genres as *boundary objects*, a concept theorized by Star and Griesemer (1989). In their study of museum artifacts, Star and Griesemer recognized that "all science requires intersectional work" (p. 392), and they identified boundary objects as necessary for coherence of information across multiple scientific worlds. They suggested that the "creation and management of boundary objects is a key process in developing and maintaining coherence across intersecting social worlds" (p. 393) and emphasized that for boundary objects to be effective, the participants making the objects must share a common purpose. But given the number of people who may then use a boundary object, the purposes may change for different people, as Russell (1997) suggested in his study of genres and activity systems. Moreover, Star and Griesemer even stated that some kinds of boundary objects, specifically those called repositories, can be used by people from various fields for their own purposes.

Star and Griesemer (1989) seemed to view boundary objects as almost benevolent; at the very least, they characterized boundary objects as able to mediate the central tensions between diverse fields to achieve coherence. Their characterization suggests that tensions reside almost naturally between different disciplines or professions and that genres that serve several different professions are able to alleviate these tensions. But when we view boundary objects in light of the professional and disciplinary successes and dominance recognized by Halpern (1992) and Detweiler and Peyton (1999), we see that boundary objects are just as likely to create and participate in dominance and tension between diverse and sometimes competing fields. As Ceccarelli (2001) has argued in her study of motivational texts of the scientific discipline, genres are far from being neutral receptacles of community values and actively participate in professional tension and dominance. And as Bawarshi (2000) stated, genres constitute the social situation, so if professions compete for dominance and power, then genres, as active participants in that situation, help create that competition. Thus, boundary genres—that is, genres functioning as boundary objects—may actively participate in interprofessional struggles about hierarchies, dominance, and values, helping to create, mediate, and store tensions.

To clarify the ways in which boundary objects cross borders, Star and Griesemer (1989) described boundary objects as those

objects which are both plastic enough to adapt to local needs and the constraints of the several parties employing them, yet robust enough to maintain a common identity across sites. . . . They have different meanings in different social worlds, but their structure is common enough to more than one world to make them recognizable, a means of translation. (p. 393)

In this definition, Star and Griesemer (1989) emphasized a boundary object's flexibility for change and its commonality across multiple fields. To move across professions, a boundary object must have a recognizable commonality of certain elements, either of form, content, disciplinary knowledge, or method. A boundary object serves the needs of multiple sites or multiple professions by being both flexible and stable and by incorporating familiar and common elements.

Differing disciplines or professions recognize this commonality through specific strategies explicated by scholars who work on problems of disciplinarity and interdisciplinarity, such as Fuller (1993) and Journet (1997). They theorized that through such moves as translation and reflexion, disciplines experience this commonality of crossing boundaries. I add another move, distillation, through which one profession may construct its relation to another profession.

In her theory of translation, Journet (1997) suggested that the knowledge of one field may play an active role in the genres used in another field. Her "conception of boundary rhetoric suggests that in certain situations writers may sublimate differences [between disciplines] not just by articulating connections but also by recasting the knowledge claims of one discipline into the [genre] generic forms of the other" (p. 57). In other words, genres that cross disciplinary boundaries may do so by incorporating the knowledge, or content, of a discipline into the generic forms of another discipline or vice versa—the content, or knowledge, of the second discipline may be incorporated into the generic forms of the original discipline. Using a metaphor of language, Journet described this "primary sublimating strategy [as] 'translation'—transforming concepts from one perspective into the vocabulary of the other" (p. 61). Thus, the process of translation does not just convert the vocabulary of a discipline's knowledge but may also convert theories, methodologies, and forms.

Journet's (1997) study of boundary genres builds on an earlier theory of interdisciplinarity by Fuller (1993), who proposed that interdisciplinarity occurs through the interpenetration or dialecticism between differing realms of thought. In defining *interpenetration* as the processes of exporting and importing ideas, he distinguished between rhetoric that seeks to unify thought toward an action or reaction and rhetoric that seeks to accentuate differences toward building knowledge. This interpenetration can be traced through a strategy that he called "reflexion," which highlights a discipline's blind spots by emphasizing the constructedness of its identity (p. 63). For analyzing medical forms as boundary genres working within the professions of business and medicine, then, Fuller's strategy of reflexion-through which one profession can recognize and describe the constructedness of what passes for common sense or naturalness in the second profession-seems especially useful.

Another important strategy for crossing professional boundaries is distillation, in which a discipline takes key theories or ideas from another discipline and summarizes them into kernels of knowledge to use for its own purposes. This distillation occurred in Berkenkotter and Ravotas's (1997) study of psychotherapy, in which the patients' complex and contextually grounded accounts were shortened and translated into the nomenclature and codes recognized by the *Diagnostic and Statistical Manual of Mental Disorders*. For example, what patients described as "feeling sad, crying all the time, and too tired to get up," the mental health caregivers diagnosed as depression. Thus, the lengthy, descriptive patient accounts are distilled into one- or two-word diagnoses. In studying medical forms, I consider distillation along with translation and reflexion as the most influential moves in crossing professional boundaries.

Looking at boundary genres of medical practices, I see that aspects of the business profession surround and subsume the genres for patient care (and by extension, the patients) from the time patients enter the medical practice until long after they leave. Moreover, as the business profession struggles with the boundary genres involved in making claims and filling the needs of the medical practice, it translates and distills the scientific and medical knowledge it has subsumed. In crossing these professional boundaries, then, the business profession uses reflexion to uncover the professional hierarchies of medicine and its genres. This understanding of the medical profes-

sion is important because it leads to ways of understanding the tensions of health care that may affect patient care.

During the course of medical treatment, several different forms may be filled out, as many as a dozen forms for one patient's single visit to the doctor. I collected many of these forms while gathering data about health-care communication at a midsized medical practice.¹ In that research, I interviewed many health-care personnel, including business managers, clerical staff, nurses, physicians assistants, and physicians; I observed the interactions between business staff and the health-care providers; and I collected more than 100 documents, including copies of 13 forms that were involved in one patient's series of two visits to the physician. I chose to analyze five of these forms from this patient because they are common to most medical practices and show significant commitment to the professions of science and business. I also chose those five forms because they are not only typical of many forms at medical practices but also remarkable enough to reveal the distinguishable maneuvers of science and business. The remaining eight forms for that single patient were so similar in design and content as to be virtually indistinguishable from each other. Of the five forms I offer here, two of these forms were specifically written for the medical practice in which they resided, whereas three of the forms are nationally produced, including the universal billing claim form that is nationally mandated for use by all health-care practitioners. Even those forms that seem unique in the minute details are almost generic and universal in their purposes; in short, the forms are suited for the way that most medical personnel conduct their work on a daily basis.

The interdisciplinary functioning of a particular profession is best understood when analyzing several interrelated genres. Looking at this big picture brings into relief the disciplinary or professional commitments that each individual genre may have in comparison to the other genres. For example, one genre may be based primarily in a particular profession, such as business, whereas another genre may be heavily influenced by scientific values. Analyzing several of these genres together as a set helps us see the ways in which disciplinary or professional involvement occurs across the system, not just within single documents. Tracing the genre system as boundary objects allows us to see the roles and influence of the various professions: how they constrain, limit, and expand one another.

In these medical practices and in the genres of these medical sites, the business profession has the lion's share of involvement in the

paperwork and communication-related work done behind the scenes. In the practice I observed, as many as 60% of the employees worked solely on completing the business of filing claims and bills; there were always more business-related employees than health-care providers (i.e., nurses, physicians, and physician assistants) present. Much time, effort, and emphasis are placed on the business end of patient care. Despite the major role business plays in the genre system, however, few medical business communicators I interviewed reported knowing how to balance the various discourses of medicine and business as they worked to make their own boundary genres recognizable to other communities.

In the following sections, I analyze each of the five forms, describing and analyzing each one according to the strategies of translation, distillation, and reflexion.

PATIENT EXAMINATION FORM

The patient examination form (see Appendix A) is used usually for yearly well-patient checkups and rarely for visits when the patient has a medical complaint or illness. This form is much like a template, with appropriate sections reminding the physician to carefully observe and recount the data regarding various parts of the patient's body. Each section has a category heading (e.g., "Respiratory," "Neuro" [neurological], "G I" [gastrointestinal]) and several subheadings that help break down into smaller detail the data and signs. This form carefully constrains the new information by limiting it only to physical data from the patient's body. Quite scientifically, the physician analyzes, observes, and records minute data, taking careful notes about those natural signs. Thus, the patient becomes the natural data observed, analyzed, recorded, and put back together as a scientific theory of the natural world (i.e., the diagnosis and plan of treatment). Schryer (1999) has argued that this process also occurs in the form of the scientific research article, in which the author-scientist strategically places the data in the chronological past to strengthen the sense that the natural phenomena have become provable and recordable scientific data. In much the same way, the health-care giver records the phenomena relating to the patient's body, casting the patient less as a person than as scientific data.

Bawarshi (2000) too described the technique of patients becoming scientific phenomena to be recorded by the scientist-physician. He

noted in his analysis of the patient medical history form that the genre works to “construct the patient as an embodied object” (p. 353) by isolating the patient’s physical symptoms from the patient’s cognizance. “So powerful is the socializing power of genre in identity formation,” Bawarshi argued, that patients “accept and act out [their] roles” (p. 354) as passive, embodied natural phenomenon. This form along with other scientifically influenced forms, such as the patient visit form (described in the next section), work to cast the physician in the role of the objective and authoritative scientist, minimizing the patient’s input, agency, and humanity.

By casting the physician in the role of the scientist and the patient in the role of the scientific data, the patient examination form uses the process of translation to cross the boundaries between science and medicine. According to Journet (1997) and Fuller (1993), translation involves the minimization of differences between two or more disciplines so that ideas of one discipline are exported to another in which they seem to be a natural extension of that discipline’s own ideas. This patient examination form converts the scientific methods of gathering data and creating theories based on that data into the medical process of recording signs and creating a diagnosis based on those signs. Furthermore, the form converts the role of patient into that of scientific, observable data and signs. Through these conversions, the form translates scientific concepts into the vocabulary of medicine. This form, however, does not cross into the profession of business; it does not work to connect the three fields of science, medicine, and business. Instead, the form functions to connect the two fields of science and medicine to make medicine look scientific.

PATIENT VISIT FORM

The use of the patient visit form (see Appendix B) begins when the patient comes to the medical practice to seek treatment for a specific ailment. On meeting the patient, the medical staff, specifically either a nurse or a medical assistant, records the patient’s name, age, blood pressure, respirations per minute, known allergies, and chief complaint. In this particular case, the top section of the form contains a record of the date and time of the patient’s first visit relating to this specific ailment. The next section contains a record of the patient’s second visit beginning with the patient’s blood pressure and the rea-

son for the patient's visit (to discuss the results of an earlier radiology exam and lab results), all of which were recorded by the nurse who escorted him to the examination room. Once the patient is inside the examination room, this patient visit form is placed in the patient's file outside the door of the examination room. When entering the room, the physician picks up the file and begins to record observations of the patient, the diagnoses, and the treatment plan. In this particular case, the physician wrote his account of the visit ("lengthy discussion in re. labs, meds, diet, exercise, smoking") along with the summary of the laboratory test results. Then, he entered his impressions, or initial diagnoses, followed last by his plan for treatment. More than the patient examination form, this patient visit form has much open space, allowing greater flexibility for recording patient and physician thoughts. In short, on the patient visit form, the physicians record the patient's symptoms and their own observations, in this case the radiology and the laboratory test results.

When the physician investigates the patient's symptoms and supports those symptoms with empirical evidence, such as test results, these symptoms become medical signs and are thus accorded status as official signs for diagnosis and treatment. Even those symptoms based on the patient's own empirical evidence are not considered medical signs until the physician investigates the symptoms and interprets the data from the investigation. The data from the doctor's investigation, which include the support of empirical evidence, provide the official evidence of the illness.

Thus, the doctor has the choice and responsibility in determining the patient's symptoms as an illness, and the doctor is obligated to write down the signs according to laws that prohibit the billing of treatments for fraudulent diseases. To be billable to health insurance companies, all illnesses must have their genesis recorded in writing in the genre of the patient chart. Even though the patient's symptoms and physician-interpreted signs precede the writing, these symptoms and signs do not constitute the disease. Rather, when physicians write them down in the patient chart, these signs become provable and billable diseases. Therefore, because the physician's choice of diagnosis indicates what codes are billable, the physician's agency—choice and responsibility—in reading the patient becomes the foundation of most medical practice–health insurance interactions.

This genre, the patient chart, necessarily incorporates the simultaneity of time and space. The physician and the patient are almost

always simultaneously present, a necessity of face-to-face communication. Such simultaneity, as Schryer (1999) argued, is a necessary element for "events and participants to acquire meaning" (p. 82). In such ways as reading the signs of the patient or of the illness, the physician casts the patient into the passive role of observed nature, much like that of observed nature that Schryer noted in her study of the scientific experimental article. As such, the physician, as author of the specific details of the specific treatment visit, controls the temporal representations of the visit by making static the visit itself and by controlling the future actions of treatment, as noted in the treatment plan. Furthermore, the physician as the author controls the spatial representations of the patient both by requiring that the patient be present and by breaking that physicality of the presence into readable parts (e.g., gastrointestinal observations, cardiovascular signs, and ear, nose, and throat readings). According to Schryer, the simultaneity of participants involves the opportunity for a certain degree of dialogism, which may be more or less involved. For example, the physician and patient may converse to arrive at a name for the illness, or the physician may work only with the signs of the illness, virtually backgrounding the physical presence of the patient and foregrounding instead the physicality and spatiality of the signs of illness by emphasizing the nature of data scientifically observed and recounted.

Again similar to the patient examination form, the patient visit form crosses the boundaries of science and medicine by using the process of translation. Similar to the patient examination form, it converts the role of the physician into the role of the scientist who gathers data and formulates theories about the natural phenomena and translates the role of the patient into the data or phenomena. The form translates the physician's and patient's roles into those of scientist and natural phenomena. As Journet (1997) and Fuller (1993) described translation, the ideas of one profession, in this case science, become the de facto ideas or vocabulary of another profession, in this case medicine. And when translation is most effective, the conversion is so familiar that it seems to be a natural occurrence. Moreover, the form also crosses the boundaries of science and medicine by distilling that scientific information into small kernels of medical data: The information about the patient (a complex and unique being) becomes simplified and distilled into small bits of medical data about laboratory test results, radiological examinations, blood pressure readings, and diagnostic theories. The role of the patient is broken down into scien-

tifically observable small pieces of data and is reassembled into the theories of diagnosis that will enable the physician to act or treat. Again similar to the patient examination form, the patient visit form does not cross the boundaries of the business profession; it does not cast the physician as anything other than a pseudoscientist. Instead, the form stably resides in a pseudoscientific profession, functioning scientifically in the medical profession.

DIAGNOSIS FORM AND INSURANCE FORM

The diagnosis form and the insurance form (shown together in Appendix C) are not hard-copy forms but usually appear only on the computer screen as part of a patient's information and files. The first form, the diagnosis form, records the patient's insurance information and dates of service for a specific illness so that for each visit and illness, a different form would be filled out. The form helps categorize and group together specific medical and treatment information related to a particular insurance claim. This information is used only to determine insurance eligibility for that particular illness or injury. For example, if a patient came to see the doctor because of minor injuries sustained in an automobile accident, the dates of service pertaining to that particular injury would be recorded, the form would be flagged as "Accident: auto" related, and then the claim would be filed with the automobile insurance company. In contrast, if the same patient were to see the doctor a week later for an illness unrelated to the automobile accident, say an inner-ear infection, this same form would be used, but different information would be entered, and the claim would be filed with the patient's health insurance. The only real medical information on the form is that of the diagnosis code, dates of old symptoms, and dates and degrees of disability. The rest of the information is directly business related, and even the medical information is entered only as it pertains to filing the business claim.

The second form, the insurance form, again a computerized form, aligns the medical treatment and diagnosis information with the billing information specific to that date of treatment. A separate form is filled out each time the patient is treated at the medical office or by the physician in a hospital setting. On this form, although much of the information seems medical in terms of the patient's physical treat-

ment and diagnosis, the information is entered for the sole purpose of matching it to the appropriate codes and charges for billing purposes.

These two forms, although brief in nature, cross the boundaries between science, medicine, and business by using translation. The medical profession exports its knowledge base or methodology to the business profession in which such knowledge and methods seem almost natural, and the differences between those two fields are sublimated or minimized. In other words, the medical inductive method of observing the scientific data about the patient and formulating theories of diagnoses becomes the same method that the business profession will use to make a claim about the treatment occurrences. And this process of translation (i.e., translating methodology) allows the method to seem like a natural extension to routine business procedures. Even though the majority of the paperwork, genres, personnel, and work hours are devoted to the business end of patient care, the genres devoted to medical purposes are much more likely to be saved in hard copy because these medical genres, or charts, document the existence of the patient symptoms and treatment by the physician. These medical charts become, in essence, the scientific proof for the overall business argument: that treatment was done and therefore payment should be rendered. As such, the business profession translates the scientific method of gathering empirical evidence to make a theory of diagnosis into its own purposes of receiving payment for a claim. By translating the facts of treatment and diagnosis into a business code, the business profession makes the argument that services were performed, and by keeping in hard copy or microfilm the treatment forms, the business profession provides factual proof for its claim. Thus, the business profession successfully subsumes the medical elements, making itself more stable and secure. It uses the necessary information from the medical and scientific professions to strengthen itself and conduct its role. And it uses only the information that it needs, which includes the dates of service, illness or injury, treatments performed, place of service, and diagnosis, and ignores other medical information, such as patient symptoms, theories of various applicable diagnoses, and alternative treatments.

BILLING CLAIM

The billing claim form (see Appendix D) chronologically follows that of the patient examination and patient visit forms although sev-

eral different kinds of written communication may occur before it, such as insurance and billing information forms sometimes filled out by the patients. But a simplicity of the chronologic process does not necessarily align with an equally spatial simplicity; the physical space of writing the billing claim form may be quite complex. The form is usually filled in, or written, in a different area of the medical practice by different people (the billing clerks rather than the physician), or the forms may be completed in a different office altogether because many medical offices contract their billing to a billing firm rather than hiring their own billing clerks. Thus this form does not incorporate a simultaneity of time and space, both of the physician and the patient, as do the patient visit and examination forms.

In this form, much of the information is taken out of a hierarchy of temporally sequential events, such as patient symptoms, followed by physician-read signs, followed by diagnoses, followed by a plan of treatment. Instead, the information is relegated to nonhierarchical blocks of information. The temporal events of the patient's medical treatment visit are noted only in column 24-A (see Appendix D), and the physical placement of that treatment is noted only in column 24-B and block 32. This genre assumes the simultaneity of the physician and patient in the treatment but does not require it as a necessary component in writing this form. The billing form also differs from the other four forms in the way it gives time a broader context by connecting the date of treatment to other dates of service for previous illnesses. Likewise, the place or space of treatment is connected to other places of service, such as hospitals and external labs. In such ways, the immediacy and physical presence of the medical treatment is de-emphasized, placed into a differently formulated hierarchy, and made to seem more complex, more contextualized, and less static. But the reformulation of a different hierarchy does not negate the temporal and spatial elements; it just places other elements at levels of importance. For example, this form places a greater emphasis on the billing information, such as the patient's (or the insured's) address, insurance policy number, place of employment, and secondary insurance information. In this way, the placement or spatiality of the patient's place of abode and place of employment—information that is used to determine billing practices—is at a greater level of prominence than the patient's place of service.

More significant, the role of the physician is placed in a role equal to or less important than that of the patient. Here, the information regarding the physician and the treatment rendered is relegated to

approximately 10 boxes rather than half of the form. And the information about the physician is placed in the same role as that of the patient. They both become data, or facts, about the treatment rather than active participants in the genre. In the patient visit form, the presence of both the physician and the patient played significant roles in the genre, with the physician playing the role of author and the patient playing the role of observed nature. In the billing claim form, however, both the physician and the patient become the data, placed into the passive role of nature observed, and the business information and the treatment play center stage. Whereas the business information takes up most of the form, the medical information, recognizable from the patient visit and examination forms in which the physician reads signs and incorporates methodologies to determine diagnoses, is distilled to a few small blocks, coded down from the complexity of the patient's life and body. As such, distillation occurs in the billing claim form as this form crosses the boundaries between science and business, borrowing the scientific facts of the treatment and diagnosis and distilling those facts into business data.

DISTILLATION, TRANSLATION, AND REFLEXION

These business-type genres (the diagnosis and insurance forms along with the billing claim form) require the use of diagnostic or treatment codes drawn from a standardized set of codes published by the American Medical Association, which lists all the recognized treatment and diagnostic codes used by medical practices, clinics, hospitals, and health insurers. These codes form the basis of acceptable billing and payment practices. The medical information involved in patient care, gathered by physician observations of the signs of the patient's physical presence, becomes distilled into a set of treatment and diagnostic codes. All the recorded observations of signs and symptoms, all the data recorded, mostly from the patient visit forms, patient examination forms, and radiological readings, are condensed into a single code as the diagnosis of the patient's illness.

In their study, Berkenkotter and Ravotas (1997) noted that the use of diagnostic codes is salient because the codes "operate to translate the clients emic categories into the professional psychiatric categorizations" (p. 269). Their study clearly shows that the translation pro-

cess involves distilling the patient's words into short diagnostic codes recognized by the psychiatric community. They state that "somewhere along the way, however, the client's richly descriptive narrative is lost, the result of the therapist's use of rhetorical and linguistic strategies" (p. 271). The medical information that dominates the patient examination and visit forms in making an argument or theory for diagnosis and plans of appropriate treatment plays only a minor role in other forms, such as the billing form. In these other forms, the medical theories and even the Latinate vocabulary for signs and diagnoses are distilled into small kernels of numerically coded information, recognized and used for business purposes by employees in the business professions, such as billing clerks and insurance company personnel.

Finally, by translating and distilling the medical method and information, the whole genre system of medical forms participates in *reflexion*, which Fuller (1993) defined as highlighting the other discipline's "blind spots" by emphasizing the constructedness of that field's identity. According to Fuller, this process is both reflexive and reflective. In other words, "knowledge is *in* the same world that it is *about*" (p. 62). Reflexion, then, for interdisciplinary maneuvers, is the act of using a second discipline to create an image of the first discipline. This reflexion, almost invisible in individual genres, is brought into focus when viewing interrelated genres. Reflexion is accomplished when the business-related practices show the constructedness of medicine and, by doing so, call attention to the very nature and authority of the physician. When the physician casts the patient in the passive role of nature observed and recorded, that is called medicine. Yet when the business managers cast the physician into the role of nature recorded to make an argument that treatment was performed, the physician is no longer the authority of nature but the passive participant, one whose authority is undermined and distilled into a single standardized code. In this reflexion process, the role of medicine is subsumed and de-emphasized as a constructed, not a natural, phenomenon. When the business profession borrows, or translates and distills, many of the same strategies that the medical profession has borrowed from the scientific profession, it reflexively shows the unnatural, or constructed, profession of medicine. And this emphasis on the constructedness of the medical profession brings into focus the competition between science, medicine, and business as each profession competes for power within the medical profession.

Some would argue that this competition came about through the medical field's willing surrender of business-related activities to more distant billing and claims professionals. Later, this strategy called attention to the physicians' lack of inherent natural authority, what Starr (1982) argued is "a profound loss of autonomy" (p. 446) entailed through the corporatization of medicine. As such, the business profession, although not dominating all the genres in the genre system and still clearly dependent on the medical information, surrounds and subsumes the medical profession by drawing from its method and data. And by drawing itself into closer and closer proximity to the medical profession, the business profession gains a certain amount of dominance, control, and authority.

CONCLUSION

We need to consider the texts of health care through an interdisciplinary or interprofessional lens to understand how these forms operate as boundary genres. Through this lens, we see that the levels of disciplinary or professional commitment vary from form to form: In some forms, recording the medical information is the sole objective whereas in other forms, that medical information is relatively minor in relation to the business objective of filing the billing claim. As boundary genres, these forms show a commonality of knowledge, vocabulary, and methodology across professions, even though they may be used for differing audiences, in differing formats, and for a variety of purposes. And as boundary genres, these forms are heteroglossic in voice and authorship, flexible enough for a variety of purposes and yet stable enough to be recognized and used in both local as well as national communities. In addition, considering theories of genre and disciplinarity as those theories that are called into specific practices strengthens our notions of professionalism and highlights the constructedness of various disciplines and professions, enlarging our knowledge of disciplines and our view of genres and discourses. Such an interprofessional examination shows us that professional boundaries are crossed and recrossed, made more distinct in some

places and almost indiscernible in others, through processes of translation, distillation, and reflexion. Through these processes, the tensions between professions (e.g., the medical practitioners resenting their lack of authority) as well as the areas of similarity are highlighted.

Finally, health care at present is a site ripe for increasing examination; its importance to all citizens suggests that we need to expand our scholarship in that area by exploring, describing, and interpreting its texts so that we can more greatly understand how those communicative practices affect the people of its various communities. As Fuller (1993) stated, "certain sorts of problems—increasingly those of general public interest—are not adequately addressed by the resources of particular disciplines, but rather require that practitioners of several such disciplines organize themselves in novel settings and adopt new ways of regarding their work and co-workers." (p. 33) As scholars of professional communication, we are well suited to enter that discussion about the communicative practices of medicine.

APPENDIX A

Patient Examination Form

History
EXAM: CONSTITUTIONAL
General Appearance
 Clean, well-groomed, no obvious abnormalities
HEENT
 Eyes: clear, no conjunctival injection, pupils equal, reactive, extraocular muscles intact
 Ears: no tenderness, no discharge
 Nose: no tenderness, no discharge
 Throat: no tenderness, no exudate, tonsils not enlarged
Neck
 No lymphadenopathy, no thyroid enlargement, no carotid bruits
Chest
 No tenderness, no crackles, no wheezes, no hyperinflation
Abdominal
 No tenderness, no masses, no organomegaly
Extremities
 No deformities, no edema, no clubbing, no cyanosis
Neurological
 No focal deficits, no abnormal reflexes, no Babinski sign
Psychiatric
 No mood or affect changes, no delusions, no hallucinations
Review of Systems
 All systems reviewed, no significant findings

Diagnosis
 1. Chronic fatigue syndrome
 2. Anxiety disorder
 3. Depression
 4. Post-traumatic stress disorder
 Other: none

Management
 1. Supportive care
 2. Cognitive behavioral therapy
 3. Medication (if indicated)
 4. Regular exercise
 5. Adequate sleep

Follow-up
 1. 4 weeks
 2. 8 weeks
 3. 12 weeks
 4. 16 weeks
 5. 20 weeks

APPENDIX B

Patient Visit Form

[illegible]

APPENDIX C Diagnosis Form and Insurance Form

Patient : F J. C. Use Unit :
 Hold Insurance Bill (Y/N/I) : N DIAGNOSIS 562.10
 Date 1st Consulted : 09/10/99 Prior Authorization # :
 Facility # : 2 [LINCOLN T. MEDICA] Inpatient (Y/N) :
 Ref Dr : F [R. B. F. M.D.]
 Related to Employment (Y/N) : Old Symptoms (Y/N) :
 ACCIDENT : (Auto, Other, None) : Emergency (Y/N) :
 Date 1st Symptom : 09/10/99
 Date Resumed Work : Special :
 Dates of Disability (Partial) : to
 Dates of Disability (Total) : to
 Dates of Similar Illness : to
 Dates of Hospitalization : to
 Outside Lab (Y/N) ? : EPSDT (Y/N) ? :
 Lab Charges : 0.00 Family Planning (Y/N) ? :
 Lab Facility # : 0 []

(Modify, (Delete or (Copy :)

Patient #: 6050.0 Name : G. F. Dept. : 0
 Voucher #: Dr # : 2 R. M.D., HRNR Location : 2
 Billed

Date	P.O.S.	Procedure	Modifier	Diagn 1-4	Units	Charges	T.O.S.
09/10/99	5	54280			1.00	\$ 216.00	4

 [X-Ray, Colon Air Contrast W/BARIUM] 562.10 DIVERTICULOSIS, DIVERTICULUM
 Receipt and Adjusts : 0.00
 Comment :
 Inst Price: 3.0 Assign/Est: Y Sec#: 0.0 Alld: 2 FMC Billable
 Insurance:
 Plan: Plan Name Effective Date Assign Note
 3.0 BC/BS OF KY - Y PRIM

Enter (Modify or (Delete :)

Billing Claim Form

[illegible]

NOTE

1. I conducted research at a midsized family medical practice in a small town in a midwestern state. During my study, I collected several documents pertaining to patient visits. Each document had the identification of the patient removed before I was given the document so that patient confidentiality was maintained. The study was approved by the Human Studies Research Board of the University of Louisville.

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